Algebraic tunneling for the restricted fractional Laplacian on distant domains

Leslie P. Polzer

Independent Researcher

polzer@fastmail.com

September 5, 2026

Abstract. We study the restricted fractional Laplacian $(-\Delta)^s$, $0 < s < 1$, on distant translates of a bounded Lipschitz open set $D \subset \mathbb{R}^d$. Let μ_1 be the one-well ground-state eigenvalue, let ϕ_1 be its positive L^2-normalized eigenfunction, and set $m_1 = \int_D \phi_1$. For $\Omega_L = (D - Le/2) \cup (D + Le/2)$ with a fixed unit vector e, we prove, as $L \to \infty$,

$$\lambda_{1,2}(\Omega_L) = \mu_1 \mp c_{d,s} m_1^2 L^{-d-2s} + O(L^{-d-2s-2} + L^{-2d-4s}),$$

where the minus sign corresponds to λ_1 and $c_{d,s}$ is the standard singular-integral normalization. When $D = -D$, we also give an exact decomposition into symmetric and antisymmetric sectors under central inversion and an explicit separation threshold. For fixed $N \geq 2$ and distinct sites $a_1, \ldots, a_N$, let $\Omega_{\mathbf{a},L} = \bigcup_{j=1}^{N}(D + La_j)$. Its first N eigenvalues satisfy $\lambda_k(\Omega_{\mathbf{a},L}) = \mu_1 + L^{-d-2s}\theta_k + O(L^{-d-2s-2} + L^{-2d-4s})$, where $\theta_1 \leq \cdots \leq \theta_N$ are the eigenvalues of the matrix with zero diagonal and off-diagonal entries $-c_{d,s} m_1^2 |a_i - a_j|^{-d-2s}$. The estimates follow from spectral-gap bounds and Taylor expansion with a quadratic remainder. For two equal balls, the expansion for λ_2 gives the leading rate in the nonlocal Hong–Krahn–Szegő minimizing sequence. Galerkin computations using exact cell integrals illustrate the two- and three-well asymptotics at fixed mesh.

Keywords. restricted fractional Laplacian; eigenvalue splitting; nonlocal double wells; spectral perturbation; shape optimization.

2020 Mathematics Subject Classification. 35P15; 35R11.

1. Introduction

The restricted fractional Laplacian couples disjoint components through its integral kernel, unlike the local Dirichlet Laplacian, whose spectrum on a disconnected union is exactly the multiset union of the component spectra. Here tunneling refers to the splitting of the one-well ground-state level caused by the nonlocal interaction between distant copies of a reference well. We consider identically oriented translates of a common well; the splitting decays algebraically with their separation.

Theorem 3.3 gives explicit bounds for the two eigenvalue shifts in the centrally symmetric case. Theorem 4.6 approximates the first N eigenvalues for a fixed family of translates by the spectrum of an explicit interaction matrix, with error bounds uniform over those N eigenvalues. Its two-well specialization, Corollary 4.7, gives the doublet asymptotic without central symmetry. For two equal balls, Corollary 3.4 identifies the leading rate along the Hong–Krahn–Szegő minimizing sequence. Section 5 illustrates the two- and three-well asymptotics by Galerkin computations at fixed mesh.

Central symmetry permits an exact decomposition into the symmetric and antisymmetric sector operators $A_D \pm B_L$. For the general family of translates, the linear term in the kernel expansion cancels after projection onto the one-well ground states. This gives the same remainder order without symmetry. The proofs use the one-well spectral facts recalled in Section 2, together with elementary perturbation estimates.

Brasco and Parini proved that the fractional Hong–Krahn–Szegő bound for the second eigenvalue is strict and that two equal balls form a minimizing sequence when their mutual distance tends to infinity [BP16, Theorem 6.2]. Goel and Sreenadh obtained an analogue for a local p-Laplacian coupled to a nonlocal p-Laplacian with a compactly supported kernel [GS19, Theorem 1.4]; the latter interaction is finite-range rather than algebraic. Biagi, Dipierro, Valdinoci, and Vecchi proved a mixed local/nonlocal analogue [BDVV23, Theorem 1.1]. Parini and Salort proved a fractional concentration–compactness trichotomy for bounded Sobolev sequences and transferred its dichotomy to quasi-open domains through operator-norm resolvent approximation [PS20, Theorems 1.1 and 1.2]. For two one-dimensional patches, Léculier and Roquejoffre proved monotonicity and continuity of the principal eigenvalue throughout the separation range, including contact [LR23, Theorem 1]. These results describe the qualitative dependence on separation and the limiting shape-optimization behavior.

The influence of mutual position on the restricted fractional spectrum is also emphasized by Abatangelo, Felli, and Noris [AFN20, arXiv v2, Introduction, p. 3], in a study of small fractional-capacity perturbations. In the shape-optimization setting, Zahl isolated the exact cross-energy under translations and proposed a signed point-mass interaction with kernel $|x-y|^{-d-2s}$ [Zah26, Section 10, equations (10.3) and (10.7), and Conjecture 10.9]. Wu subsequently proved that this discrete signed-mass model has no local minima, settling that conjecture [Wu26, Introduction]. Dipierro, Proietti Lippi, Sportelli, and Valdinoci studied disconnected domains for generalized eigenproblems having a superposition of lower-order operators on the right-hand side [DPLSV26, Theorems 1.4–1.7].

Zahl's discrete energy anticipates the matrix in (4.9). Here we derive that matrix from the standard restricted fractional Laplacian, controlling the interaction in operator norm and the contribution of complementary modes through the one-well spectral gap. We are not aware of an earlier result giving either the two-component coefficient with a quantified remainder or this multi-well spectral reduction.

2. The one-well operator and exact two-well reduction

Fix $d \geq 1$ and $0 < s < 1$, and put

$$\kappa = d + 2s.$$

For an open set $G \subset \mathbb{R}^d$, let

$$\widetilde{H}^s(G) = \{u \in H^s(\mathbb{R}^d) : u = 0 \text{ a.e. on } \mathbb{R}^d \setminus G\}.$$

We use the Gagliardo seminorm

$$[u]^2_{H^s(\mathbb{R}^d)} \;=\; \iint_{\mathbb{R}^d\times\mathbb{R}^d} \frac{|u(x)-u(y)|^2}{|x-y|^{d+2s}}\,dx\,dy.$$

For bounded Lipschitz G, this space agrees with the completion of $C_c^\infty(G)$ in the global $H^s(\mathbb{R}^d)$ norm [DFW21, Section 2, equation (2.1)]. The zero-exterior quadratic form is

$$\mathcal{Q}_G[u] \;=\; \frac{c_{d,s}}{2}\iint_{\mathbb{R}^d\times\mathbb{R}^d} \frac{|u(x)-u(y)|^2}{|x-y|^{\kappa}}\,dx\,dy, \qquad u\,\in\,\widetilde{H}^s(G), \tag{2.1}$$

where $c_{d,s} \,=\, 4^s s\,\Gamma(d/2+s)/(\pi^{d/2}\Gamma(1-s))$ is the standard normalization in the singular-integral definition of $(-\Delta)^s$. Brasco and Parini use coefficient 2 in place of $c_{d,s}$ in the singular-integral operator, so their eigenvalues are multiplied here by $c_{d,s}/2$; the positivity, simplicity, ordering, and shape-optimization statements cited below are unchanged by this fixed positive rescaling [BP16, arXiv v2, Section 1.1, p. 2]. The form norm $\big(\mathcal{Q}_G[u]+\|u\|^2_{L^2(G)}\big)^{1/2}$ is equivalent to the norm inherited from $H^s(\mathbb{R}^d)$ on $\widetilde{H}^s(G)$. Thus (2.1) is a densely defined closed form, and we denote its associated self-adjoint operator on $L^2(G)$ by A_G. We write $\lambda_1(G)\,\le\,\lambda_2(G)\,\le\,\cdots$ for the eigenvalues of A_G, repeated according to multiplicity. For an abstract self-adjoint operator T with compact resolvent, $\lambda_k(T)$ has the analogous meaning.

Let $D\,\subset\,B_R(0)$ be a bounded Lipschitz open set satisfying $D\,=\,-D$. We retain Lipschitz regularity in order to invoke the extension property and compactness of the form embedding from [DNPV12, Theorems 5.4 and 7.1]. Hence A_D has compact resolvent. Its first eigenvalue is simple and has a strictly positive normalized eigenfunction by [BP16, Theorem 2.8], specialized to the linear case. Put

$$\mu_1\;=\;\lambda_1(D), \qquad \mu_2\;=\;\lambda_2(D), \qquad g\;=\;\mu_2-\mu_1\;>\;0.$$

Fix the positive normalized ground-state eigenfunction and set

$$A_D\phi_1\;=\;\mu_1\phi_1, \qquad \|\phi_1\|_{L^2(D)}\;=\;1, \qquad m_1\;=\;\int_D \phi_1(x)\,dx\;>\;0. \tag{2.2}$$

Simplicity and invariance under central inversion show that $\phi_1(-x)\,=\,\pm\phi_1(x)$; positivity fixes the sign, so $\phi_1(-x)\,=\,\phi_1(x)$.

Fix a unit vector $e\,\in\,\mathbb{R}^d$. For $L\,>\,2R$, define

$$D_L^-\;=\;D-\frac{L}{2}e, \qquad D_L^+\;=\;D+\frac{L}{2}e, \qquad \Omega_L\;=\;D_L^-\cup D_L^+.$$

On $L^2(D)$ define the bounded integral operator

$$(B_Lv)(x)\;=\;-c_{d,s}\int_D \frac{v(y)}{|Le-x-y|^{\kappa}}\,dy. \tag{2.3}$$

Its kernel is real and symmetric, so B_L is self-adjoint.

Lemma 2.1 *(Exact reduction by central inversion)***.** *The operator A_{Ω_L} is unitarily equivalent to*

$$\begin{pmatrix} A_D & B_L \\ B_L & A_D \end{pmatrix} \quad \text{on } L^2(D) \oplus L^2(D),$$

and hence to

$$(A_D + B_L) \oplus (A_D - B_L). \tag{2.4}$$

Proof. For $f \in \widetilde{H}^s(\Omega_L)$ set

$$u(x) = f\left(x - \frac{L}{2}e\right), \qquad v(x) = f\left(-x + \frac{L}{2}e\right), \qquad x \in D.$$

This defines a unitary map from $L^2(\Omega_L)$ to $L^2(D) \oplus L^2(D)$. Translation invariance and invariance under central inversion identify the two diagonal parts of the form with $\mathcal{Q}_D[u]$ and $\mathcal{Q}_D[v]$.

To identify the form domains, fix $\chi \in C_c^\infty(\mathbb{R}^d)$ and $w \in H^s(\mathbb{R}^d)$. Then

$$\begin{aligned}[]
[\chi w]^2_{H^s(\mathbb{R}^d)} &\le 2\|\chi\|^2_{L^\infty} [w]^2_{H^s(\mathbb{R}^d)} \\
&\qquad + 2\int_{\mathbb{R}^d} |w(y)|^2 \int_{\mathbb{R}^d} \frac{|\chi(x) - \chi(y)|^2}{|x-y|^{d+2s}}\, dx\, dy.
\end{aligned}$$

The inner integral is bounded uniformly in y: use $|\chi(x) - \chi(y)| \le \|\nabla \chi\|_{L^\infty} |x-y|$ when $|x-y| \le 1$ and $|\chi(x) - \chi(y)| \le 2\|\chi\|_{L^\infty}$ otherwise. The two radial bounds are integrable because $s < 1$ and $s > 0$, respectively. Consequently multiplication by χ is bounded on $H^s(\mathbb{R}^d)$.

Positive separation permits cutoffs $\chi_\pm$ that equal one on $D_L^\pm$ and vanish near the other component. For $f \in \widetilde{H}^s(\Omega_L)$, the functions $\chi_- f$ and $\chi_+ f$ therefore belong to $H^s(\mathbb{R}^d)$ and, after the above changes of variables, give $u, v \in \widetilde{H}^s(D)$. Conversely, translated and reflected zero extensions of any $u, v \in \widetilde{H}^s(D)$ belong to $H^s(\mathbb{R}^d)$, and their sum is supported in Ω_L. Hence the unitary identifies $\widetilde{H}^s(\Omega_L)$ with $\widetilde{H}^s(D) \oplus \widetilde{H}^s(D)$.

The cross-kernel is bounded because of positive separation. Each separate zero-exterior form already includes the squared-modulus terms integrated against the kernel over the other component. Thus, after the above changes of variables, expanding the square in (2.1) yields the exact identity

$$\begin{aligned}
&\mathcal{Q}_{\Omega_L}[f] - \mathcal{Q}_D[u] - \mathcal{Q}_D[v] \\
&\quad = c_{d,s} \iint_{D\times D} \frac{|u(x) - v(y)|^2 - |u(x)|^2 - |v(y)|^2}{|Le - x - y|^\kappa}\, dx\, dy \\
&\quad = -2c_{d,s} \operatorname{Re} \iint_{D\times D} \frac{u(x)\overline{v(y)}}{|Le - x - y|^\kappa}\, dx\, dy = 2\operatorname{Re}\langle u, B_L v\rangle.
\end{aligned}$$

The square terms therefore cancel, leaving no additional diagonal contribution. This proves the block representation in the sense of closed forms. Since B_L is bounded, it also gives the stated operator representation. The unitary change of variables

$$(u, v) \longmapsto 2^{-1/2}(u + v, u - v)$$

yields (2.4). ■

3. The doublet splitting

We first record the elementary perturbation estimate used for the two branches.

Lemma 3.1 *(Ground-state perturbation bound).* *Let A be self-adjoint with compact resolvent, simple lowest eigenvalue μ, and normalized ground state ϕ. Define the spectral gap by $g = \lambda_2(A) - \mu > 0$. Let W be bounded and self-adjoint, put $b = \|W\|$ and $w = \langle \phi, W\phi\rangle$, and suppose $b \le g/4$. Then*

$$-\frac{2b^2}{g} \le \lambda_1(A+W) - (\mu + w) \le 0. \tag{3.1}$$

Proof. Let $\mathfrak{a}$ be the closed quadratic form associated with A. The upper bound follows by testing the Rayleigh quotient with ϕ. For a normalized vector $\psi = a\phi + \eta$ in the form domain of $\mathfrak{a}$, where $\eta \perp \phi$ and $t = \|\eta\|$, the spectral gap gives

$$\mathfrak{a}[\psi] \ge \mu + gt^2.$$

Since $|a|^2 = 1 - t^2$, self-adjointness gives

$$\langle \psi, W\psi\rangle - w = -t^2 w + 2\,\mathrm{Re}\langle a\phi, W\eta\rangle + \langle \eta, W\eta\rangle.$$

Using $|w| \le b$, $|a| \le 1$, and $\|W\| = b$,

$$\begin{aligned}
\mathfrak{a}[\psi] + \langle \psi, W\psi\rangle - (\mu + w) &\ge gt^2 - 2bt^2 - 2bt \\
&= (g-2b)t^2 - 2bt \\
&= (g-2b)\left(t - \frac{b}{g-2b}\right)^2 - \frac{b^2}{g-2b} \\
&\ge -\frac{b^2}{g-2b} \ge -\frac{2b^2}{g}.
\end{aligned}$$

Taking the infimum over normalized ψ proves the lower bound. ■

The next lemma extracts the interaction coefficient. In the reflected coordinates used here, evenness of the ground state removes the linear Taylor term. Section 4 obtains the same error order without symmetry from an algebraic cancellation in translated coordinates.

Lemma 3.2 *(Interaction-kernel expansion).* *Let*

$$I_L = \iint_{D\times D} \frac{\phi_1(x)\phi_1(y)}{|Le - x - y|^\kappa}\, dx\, dy.$$

For $L \ge 4R$,

$$|I_L - m_1^2 L^{-\kappa}| \le C_{\kappa,R} m_1^2 L^{-\kappa-2}, \qquad C_{\kappa,R} = 2^{\kappa+3}\kappa(\kappa+1)R^2. \tag{3.2}$$

Moreover,

$$\|B_L\| \le \beta_L, \qquad \beta_L = c_{d,s} 2^\kappa |D|\, L^{-\kappa}. \tag{3.3}$$

Proof. Set $F_L(z) = |Le - z|^{-\kappa}$. For $|z| \le 2R$ and $L \ge 4R$, the segment from 0 to z remains at distance at least $L/2$ from Le. Put $n_z = (Le - z)/|Le - z|$. Direct differentiation gives

$$\nabla^2 F_L(z) = \kappa |Le - z|^{-\kappa-2}\big((\kappa+2) n_z \otimes n_z - I\big).$$

Its radial and transverse eigenvalues are respectively $\kappa(\kappa+1)|Le - z|^{-\kappa-2}$ and $-\kappa|Le - z|^{-\kappa-2}$. Since $\kappa > 1$,

$$\|\nabla^2 F_L(z)\| = \kappa(\kappa+1)|Le - z|^{-\kappa-2}.$$

Taylor's theorem therefore yields

$$|F_L(z) - L^{-\kappa} - \kappa L^{-\kappa-1} e \cdot z| \le C_{\kappa,R} L^{-\kappa-2}$$

for $|z| \le 2R$, with the displayed value of $C_{\kappa,R}$. Because ϕ_1 is even,

$$\int_D x\phi_1(x)\,dx = 0.$$

Consequently the integral of the linear Taylor term with $z = x + y$ vanishes. Since ϕ_1 is positive, integration of the remainder gives (3.2).

Finally, $|Le - x - y| \ge L - 2R \ge L/2$. The Schur test applied to (2.3) gives

$$\|B_L\| \le c_{d,s}|D|(L - 2R)^{-\kappa} \le c_{d,s} 2^{\kappa} |D|\, L^{-\kappa},$$

which is (3.3). ■

Theorem 3.3 *(Two-well algebraic splitting)***.** *Let $D \subset B_R(0)$ be a bounded centrally symmetric Lipschitz open set, and let Ω_L be as in Section 2. Set*

$$L_0 = \max\left\{4R, \left(\frac{4c_{d,s}2^{\kappa}|D|}{g}\right)^{1/\kappa}, 2\left(\frac{2c_{d,s}4^{\kappa}|D|^2}{g m_1^2}\right)^{1/\kappa}\right\}. \tag{3.4}$$

For $L \ge L_0$, the first two eigenvalues of A_{Ω_L} satisfy

$$|\lambda_1(\Omega_L) - (\mu_1 - c_{d,s} m_1^2 L^{-\kappa})| \le c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}, \tag{3.5}$$

$$|\lambda_2(\Omega_L) - (\mu_1 + c_{d,s} m_1^2 L^{-\kappa})| \le c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}. \tag{3.6}$$

Both eigenvalues are simple, and

$$\lambda_3(\Omega_L) \ge \mu_1 + \frac{3g}{4}, \qquad \lambda_3(\Omega_L) - \lambda_2(\Omega_L) \ge \frac{g}{2}. \tag{3.7}$$

In particular, as $L \to \infty$,

$$L^{d+2s}\big(\lambda_2(\Omega_L) - \lambda_1(\Omega_L)\big) \longrightarrow 2c_{d,s} m_1^2. \tag{3.8}$$

Proof. By Lemma 2.1, the spectrum is the union of the spectra of $A_D + B_L$ and $A_D - B_L$. Put

$$t_L \;=\; \langle \phi_1, B_L\phi_1\rangle \;=\; -c_{d,s} I_L \;<\; 0.$$

For $L \geq L_0$, the second term in (3.4) gives the first inequality in

$$\beta_L \;\leq\; \frac{g}{4}, \qquad 2c_{d,s} m_1^2 (L+2R)^{-\kappa} \;>\; \frac{4\beta_L^2}{g}. \tag{3.9}$$

Indeed, $L \geq 4R$ gives $L + 2R \leq 3L/2$, while the third term in (3.4) and $3/2 < 2$ give

$$L^\kappa \;>\; \left(\frac{3}{2}\right)^\kappa \frac{2c_{d,s}4^\kappa |D|^2}{g m_1^2}.$$

Substitution of $\beta_L = c_{d,s}2^\kappa |D|\, L^{-\kappa}$ proves the strict second inequality in (3.9).

Apply Lemma 3.1 first with $W = B_L$ and then with $W = -B_L$. Let $E_+(L)$ and $E_-(L)$ denote the ground-state energies of the symmetric and antisymmetric branches, respectively. They have the form

$$E_+(L) \;=\; \mu_1 + t_L + \rho_+(L), \qquad E_-(L) \;=\; \mu_1 - t_L + \rho_-(L), \tag{3.10}$$

where

$$|\rho_\pm(L)| \;\leq\; \frac{2\beta_L^2}{g}.$$

The second eigenvalue of either branch is at least $\mu_2 - \beta_L \geq \mu_1 + 3g/4$, whereas both branch ground-state energies are at most $\mu_1 + \beta_L \leq \mu_1 + g/4$. Thus the first two eigenvalues of the full operator are the two numbers in (3.10).

Since $|Le - x - y| \leq L + 2R$ and $\phi_1 > 0$,

$$I_L \;\geq\; m_1^2 (L+2R)^{-\kappa}.$$

The second condition in (3.9) and (3.10) imply $E_+(L) < E_-(L)$. Hence $\lambda_1(\Omega_L) = E_+(L)$ and $\lambda_2(\Omega_L) = E_-(L)$. The within-branch separation above makes both eigenvalues simple. It also gives $\lambda_3(\Omega_L) \geq \mu_1 + 3g/4$, while $\lambda_2(\Omega_L) \leq \mu_1 + g/4$; this proves (3.7). Equations (3.5) and (3.6) now follow from Lemma 3.2. Subtracting the two expansions and multiplying by L^κ proves (3.8), since $\kappa > 0$. ■

Corollary 3.4 *(Rate for the distant-ball minimizing sequence).* *Let $D = B_R(0)$ and let Ω_L be the union of two radius-R balls whose centers are separated by L, so that $|\Omega_L| = 2|B_R|$. Then, as $L \to \infty$,*

$$L^{d+2s}\big(\lambda_2(\Omega_L) - \lambda_1(B_R)\big) \;\longrightarrow\; c_{d,s}\left(\int_{B_R} \phi_1(x)\,dx\right)^2.$$

Thus the minimizing sequence in [BP16, Theorem 6.2] approaches its limiting value at the explicit leading order L^{-d-2s}.

Proof. This is (3.6) with $D = B_R(0)$ and $\mu_1 = \lambda_1(B_R)$. ■

4. Finitely many wells

The two-well decomposition induced by central inversion is specific to a pair. For finitely many components we instead keep the full block operator and compare its lowest spectral cluster with its compression to the translated one-well ground states.

Throughout this section, let $D \subset B_R(0)$ be any bounded Lipschitz open set; central symmetry is not assumed. The compactness and ground-state results [DNPV12, Theorems 5.4 and 7.1] and [BP16, Theorem 2.8] apply unchanged. Write $\mu_1 < \mu_2$, $g = \mu_2 - \mu_1 > 0$, choose the positive normalized ground state $A_D\phi_1 = \mu_1\phi_1$, and set $m_1 = \int_D \phi_1(x)\,dx > 0$. Every $L \to \infty$ statement in this section keeps N and the distinct sites $\mathbf{a}$ fixed.

Definition 4.1 (Multi-well geometry)**.** Fix an integer $N \geq 2$ and distinct points $\mathbf{a} = (a_1, \dots, a_N)$ in $\mathbb{R}^d$. Set

$$\delta_{\mathbf{a}} = \min_{i\neq j}|a_i - a_j|, \qquad \Sigma_p(\mathbf{a}) = \max_{1\leq i\leq N}\sum_{j\neq i}|a_i - a_j|^{-p} \quad (p > 0). \tag{4.1}$$

For $L\delta_{\mathbf{a}} > 2R$, define

$$D_{j,L} = D + La_j, \qquad \Omega_{\mathbf{a},L} = \bigcup_{j=1}^{N} D_{j,L}. \tag{4.2}$$

The sets in this union are pairwise disjoint.

On $\mathcal{H}_N = \bigoplus_{j=1}^N L^2(D)$, let $\mathcal{A}_0 = \bigoplus_{j=1}^N A_D$. For $i \neq j$ define

$$(B_{ij,L}v)(x) = -c_{d,s}\int_D \frac{v(y)}{|L(a_i - a_j) + x - y|^\kappa}\,dy, \qquad B_{ii,L} = 0, \tag{4.3}$$

and let $\mathcal{V}_L = (B_{ij,L})_{i,j=1}^N$.

Lemma 4.2 *(Exact multi-well reduction and norm bound)***.** *If $L\delta_{\mathbf{a}} \geq 4R$, then $A_{\Omega_{\mathbf{a},L}}$ is unitarily equivalent to $\mathcal{A}_0 + \mathcal{V}_L$. Moreover,*

$$\|\mathcal{V}_L\| \leq \gamma_L, \qquad \gamma_L = c_{d,s}2^\kappa|D|\,\Sigma_\kappa(\mathbf{a})L^{-\kappa}. \tag{4.4}$$

Proof. For $f \in L^2(\Omega_{\mathbf{a},L})$, put $u_j(x) = f(x + La_j)$ for $x \in D$. This is a unitary identification with $\mathcal{H}_N$. Translation invariance gives the diagonal operator $\mathcal{A}_0$. The cutoff argument in the proof of Lemma 2.1 identifies the form domain with $\bigoplus_{j=1}^N \widetilde{H}^s(D)$, and positive separation makes every cross-kernel bounded. For $i < j$, direct expansion of the form (2.1) gives the cross term

$$-2c_{d,s}\,\mathrm{Re}\iint_{D\times D}\frac{u_i(x)\overline{u_j(y)}}{|L(a_i - a_j) + x - y|^\kappa}\,dx\,dy.$$

Since $B_{ji,L} = B_{ij,L}^*$, summing these terms proves the exact block representation in the sense of closed forms and hence as operators.

For $i \neq j$ and $x, y \in D$,

$$|L(a_i - a_j) + x - y| \;\geq\; L|a_i - a_j| - 2R \;\geq\; \frac{L}{2}|a_i - a_j|.$$

The Schur test therefore gives

$$\|B_{ij,L}\| \;\leq\; c_{d,s} 2^{\kappa} |D|\, L^{-\kappa} |a_i - a_j|^{-\kappa}. \tag{4.5}$$

For $u = (u_1, \ldots, u_N)$, set $r_j = \|u_j\|_{L^2(D)}$ and let N_L be the symmetric matrix with entries $(\mathsf{N}_L)_{ij} = \|B_{ij,L}\|$. Then

$$\|\mathcal{V}_L u\|_{\mathcal{H}_N} \;\leq\; \|\mathsf{N}_L r\|_{\ell^2} \;\leq\; \left(\max_i \sum_j (\mathsf{N}_L)_{ij} \right) \|r\|_{\ell^2}.$$

Combining this inequality with (4.5) proves (4.4). ■

We next record the finite-multiplicity perturbation estimate that replaces Lemma 3.1.

Lemma 4.3 *(Isolated spectral-cluster bound)***.** *Let A be self-adjoint with compact resolvent. Suppose its lowest eigenvalue μ has multiplicity N, let P be the corresponding spectral projection, and suppose*

$$A\big|_{P^\perp} \;\geq\; \mu + g$$

in the quadratic-form sense for some $g > 0$. Let W be bounded and self-adjoint, set $b = \|W\|$, and assume $b \leq g/4$. If $\eta_1 \leq \cdots \leq \eta_N$ are the eigenvalues of PWP on $\operatorname{ran} P$, then, for $1 \leq k \leq N$,

$$-\frac{2b^2}{g} \;\leq\; \lambda_k(A + W) - (\mu + \eta_k) \;\leq\; 0. \tag{4.6}$$

In addition,

$$\lambda_{N+1}(A + W) \;\geq\; \mu + g - b \;\geq\; \mu + \frac{3g}{4}. \tag{4.7}$$

Proof. The min–max principle applied to the span of the first k eigenvectors of PWP gives the upper bound in (4.6). For the lower bound, write $\psi = p + q$ with $p = P\psi$ and $q = (I - P)\psi$. Then

$$\begin{aligned} \langle \psi, (A + W)\psi \rangle \;&\geq\; \mu\|\psi\|^2 + \langle p, PWPp \rangle + (g - b)\|q\|^2 - 2b\|p\|\|q\| \\ &\geq\; \mu\|\psi\|^2 + \langle p, PWPp \rangle - \frac{2b^2}{g}\|p\|^2 + \left(\frac{g}{2} - b\right)\|q\|^2. \end{aligned}$$

The second line uses $2b\|p\|\|q\| \leq 2b^2 g^{-1}\|p\|^2 + (g/2)\|q\|^2$. Thus, in the form sense, $A + W$ dominates the direct sum of

$$\mu I + PWP - \frac{2b^2}{g} I \quad \text{on } \operatorname{ran} P$$

and $(\mu + g/2 - b)I$ on $P^\perp$. Since $\eta_N \leq b \leq g/4$, we have

$$\eta_N - \frac{2b^2}{g} \;\leq\; b - \frac{2b^2}{g} \;<\; \frac{g}{2} - b.$$

The strict inequality follows because the difference between the last two quantities is $(g-2b)^2/(2g) > 0$. Hence the first N eigenvalues of this comparison operator are the N eigenvalues of its P-block. Another application of the min–max principle gives the lower bound in (4.6). Finally, $A+W \geq A-bI$ and the min–max principle give (4.7). ∎

Let P_1 be the orthogonal projection in $\mathcal{H}_N$ onto the span of the N vectors whose jth component is ϕ_1 and whose other components vanish. In this basis the compression $P_1\mathcal{V}_LP_1$ is represented by the real symmetric matrix $T_L = (t_{ij}(L))$, where

$$t_{ii}(L) = 0, \qquad t_{ij}(L) = -c_{d,s}\iint_{D\times D}\frac{\phi_1(x)\phi_1(y)}{|L(a_i-a_j)+x-y|^\kappa}\,dx\,dy \quad (i\neq j). \tag{4.8}$$

Definition 4.4 (Effective interaction matrix)**.** Define the real symmetric $N\times N$ matrix $M_{\mathbf{a}}$ by

$$(M_{\mathbf{a}})_{ii} = 0, \qquad (M_{\mathbf{a}})_{ij} = -c_{d,s}m_1^2|a_i-a_j|^{-\kappa} \quad (i\neq j). \tag{4.9}$$

Write its ordered eigenvalues as $\theta_1 \leq \cdots \leq \theta_N$.

Lemma 4.5 *(Compression error)***.** *If $L\delta_{\mathbf{a}} \geq 4R$, then*

$$\|T_L - L^{-\kappa}M_{\mathbf{a}}\|_{\ell^2\to\ell^2} \leq \varepsilon_L, \qquad \varepsilon_L = c_{d,s}C_{\kappa,R}m_1^2\Sigma_{\kappa+2}(\mathbf{a})L^{-\kappa-2}. \tag{4.10}$$

Here $C_{\kappa,R}$ is given by (3.2).

Proof. Fix $i\neq j$, put $a = a_i - a_j$, and set $F(z) = |La+z|^{-\kappa}$. For $|z| \leq 2R$, the segment from 0 to z stays at distance at least $L|a|/2$ from $-La$. As in the proof of Lemma 3.2, Taylor's theorem gives

$$|F(z)-F(0)-\nabla F(0)\cdot z| \leq C_{\kappa,R}L^{-\kappa-2}|a|^{-\kappa-2}. \tag{4.11}$$

The linear term vanishes after integration because

$$\iint_{D\times D}\phi_1(x)\phi_1(y)(x-y)\,dx\,dy = 0.$$

This cancellation is algebraic and does not require central symmetry of D. Using $F(0) = L^{-\kappa}|a|^{-\kappa}$ in (4.8) and integrating (4.11) gives

$$|t_{ij}(L) - L^{-\kappa}(M_{\mathbf{a}})_{ij}| \leq c_{d,s}C_{\kappa,R}m_1^2L^{-\kappa-2}|a_i-a_j|^{-\kappa-2}.$$

The operator norm of a real symmetric matrix is at most its largest absolute row sum. Taking that row sum proves (4.10). ∎

Theorem 4.6 *(Finite multi-well effective interaction matrix)***.** *Use the fixed one-well data and Definitions 4.1 and 4.4, with γ_L and ε_L from (4.4) and (4.10). If*

$$L\delta_{\mathbf{a}} \,\geq\, 4R, \qquad \gamma_L \,\leq\, \frac{g}{4}, \tag{4.12}$$

then, for $1 \,\leq\, k \,\leq\, N$,

$$|\lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + L^{-\kappa}\theta_k)| \,\leq\, \varepsilon_L + \frac{2\gamma_L^2}{g}. \tag{4.13}$$

The upper edge of the cluster and the next eigenvalue satisfy

$$\lambda_N(\Omega_{\mathbf{a},L}) \,\leq\, \mu_1 + \gamma_L \,\leq\, \mu_1 + \frac{g}{4}, \qquad \lambda_{N+1}(\Omega_{\mathbf{a},L}) \,\geq\, \mu_1 + g - \gamma_L \,\geq\, \mu_1 + \frac{3g}{4}. \tag{4.14}$$

In particular, the cluster gap has the quantitative lower bound

$$\lambda_{N+1}(\Omega_{\mathbf{a},L}) - \lambda_N(\Omega_{\mathbf{a},L}) \,\geq\, g - 2\gamma_L \,\geq\, \frac{g}{2}. \tag{4.15}$$

Consequently, as $L \,\to\, \infty$, for every $1 \,\leq\, k \,\leq\, N$,

$$L^{d+2s}\big(\lambda_k(\Omega_{\mathbf{a},L}) - \mu_1\big) \,\longrightarrow\, \theta_k. \tag{4.16}$$

Proof. Lemma 4.2 identifies the full operator with $\mathcal{A}_0 + \mathcal{V}_L$. The lowest eigenvalue of $\mathcal{A}_0$ is μ_1, with multiplicity N, its spectral projection is P_1, and the rest of its spectrum lies in $[\mu_1 + g, \infty)$. Apply Lemma 4.3 with $A \,=\, \mathcal{A}_0$, $W \,=\, \mathcal{V}_L$, and $b \,=\, \|\mathcal{V}_L\| \,\leq\, \gamma_L$. If $\tau_1(L) \,\leq\, \cdots \,\leq\, \tau_N(L)$ are the eigenvalues of T_L, this gives

$$-\frac{2\gamma_L^2}{g} \,\leq\, \lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + \tau_k(L)) \,\leq\, 0. \tag{4.17}$$

The finite-dimensional min–max principle and Lemma 4.5 give

$$|\tau_k(L) - L^{-\kappa}\theta_k| \,\leq\, \varepsilon_L.$$

The triangle inequality gives (4.13). Using $\tau_N(L) \,\leq\, \|T_L\| \,\leq\, \|\mathcal{V}_L\| \,\leq\, \gamma_L$ in (4.17) gives the upper bound in (4.14); (4.7) gives the lower bound. Their difference proves (4.15). Finally, $L^\kappa\varepsilon_L \,=\, O(L^{-2})$ and $L^\kappa\gamma_L^2 \,=\, O(L^{-\kappa})$ with $\kappa \,=\, d + 2s \,>\, 0$, so multiplication of (4.13) by L^κ gives (4.16). ∎

Corollary 4.7 *(Symmetry-free two-well asymptotic)***.** *Let $D \,\subset\, B_R(0)$ be any bounded Lipschitz open set, let $a_1 \,\neq\, a_2$, and put $r \,=\, |a_1 - a_2|$. For $\Omega_L \,=\, (D + La_1) \cup (D + La_2)$, as $L \,\to\, \infty$,*

$$\begin{aligned}
\lambda_1(\Omega_L) \,&=\, \mu_1 - c_{d,s} m_1^2 r^{-\kappa} L^{-\kappa} + O\big(L^{-\kappa-2} + L^{-2\kappa}\big)\,, \\
\lambda_2(\Omega_L) \,&=\, \mu_1 + c_{d,s} m_1^2 r^{-\kappa} L^{-\kappa} + O\big(L^{-\kappa-2} + L^{-2\kappa}\big)\,.
\end{aligned}$$

In particular,

$$L^{d+2s}\big(\lambda_2(\Omega_L) - \lambda_1(\Omega_L)\big) \,\longrightarrow\, 2c_{d,s} m_1^2 r^{-d-2s}.$$

The constants implicit in the two expansions may depend on d, s, D, R, and r, but are independent of L.

Proof. For $N = 2$, the effective matrix is

$$M_{\mathbf{a}} = \begin{pmatrix} 0 & -c_{d,s}m_1^2 r^{-\kappa} \\ -c_{d,s}m_1^2 r^{-\kappa} & 0 \end{pmatrix},$$

whose ordered eigenvalues are $-c_{d,s}m_1^2 r^{-\kappa}$ and $c_{d,s}m_1^2 r^{-\kappa}$. Theorem 4.6 gives both expansions because $\varepsilon_L = O(L^{-\kappa-2})$ and $\gamma_L^2 = O(L^{-2\kappa})$. Subtracting them and using $\kappa = d + 2s$ gives the limit. ■

Corollary 4.8 *(Two-well remainder comparison)***.** *Assume in addition that $D = -D$, and take $N = 2$, $a_1 = -e/2$, and $a_2 = e/2$. For every $L \geq L_0$ from (3.4), the error expression $\varepsilon_L + 2\gamma_L^2/g$ in (4.13) is exactly the right-hand side of each bound (3.5)–(3.6).*

Proof. In this case

$$M_{\mathbf{a}} = \begin{pmatrix} 0 & -c_{d,s}m_1^2 \\ -c_{d,s}m_1^2 & 0 \end{pmatrix},$$

whose ordered eigenvalues are $-c_{d,s}m_1^2$ and $c_{d,s}m_1^2$. Moreover, $\Sigma_\kappa(\mathbf{a}) = \Sigma_{\kappa+2}(\mathbf{a}) = 1$, so $\gamma_L = \beta_L$ and $\varepsilon_L = c_{d,s}C_{\kappa,R}m_1^2 L^{-\kappa-2}$. Hence

$$\varepsilon_L + \frac{2\gamma_L^2}{g} = c_{d,s}C_{\kappa,R}m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}.$$

The condition $L \geq L_0$ also implies $L \geq 4R$ and $\gamma_L = \beta_L \leq g/4$, so the hypotheses of Theorem 4.6 hold. In the exact compression, central symmetry of D and evenness of ϕ_1 show, after the change of variables $y \mapsto -y$, that the translated two-well kernel integral equals I_L. The ordered leading terms therefore match the reflected branches in Theorem 3.3, and the displayed identity proves the claim about their remainder bounds. ■

Corollary 4.9 *(Collective ground state)***.** *For every multi-well configuration, $\theta_1 < 0$ is simple and admits an eigenvector with strictly positive entries, while $\theta_N > 0$. For every L satisfying the separation condition $L\delta_{\mathbf{a}} > 2R$ in Definition 4.1, $\lambda_1(\Omega_{\mathbf{a},L})$ is simple. Moreover, for all sufficiently large L, $\lambda_1(\Omega_{\mathbf{a},L}) < \mu_1 < \lambda_N(\Omega_{\mathbf{a},L})$.*

Proof. The off-diagonal entries of $M_{\mathbf{a}}$ are strictly negative, and its Rayleigh quotient at $(1, \ldots, 1)$ is negative, so $\theta_1 < 0$. Replacing a Rayleigh minimizer z by its componentwise absolute value cannot increase the quotient; equality forces its nonzero components to have the same sign, and the eigenvalue equation rules out a zero component. Thus the components of a ground-state eigenvector are either all strictly positive or all strictly negative. If the ground eigenspace had dimension at least two, it would contain a nonzero vector with a zero component, a contradiction. Hence θ_1 is simple. Finally, $\operatorname{tr} M_{\mathbf{a}} = 0$ gives $\theta_N > 0$. Put $r_L = \varepsilon_L + 2\gamma_L^2/g$. Theorem 4.6 gives $L^\kappa r_L \to 0$. Simplicity of $\lambda_1(\Omega_{\mathbf{a},L})$ for every admissible L follows from [BP16, Theorem 2.8], applied to the bounded open set $\Omega_{\mathbf{a},L}$. Choose L so large that

$$L^\kappa r_L < \min\{-\theta_1, \theta_N\}.$$

Then (4.13) gives $\lambda_1 < \mu_1 < \lambda_N$, which proves the remaining assertions. ■

Corollary 4.10 *(Regular-simplex cluster)*. *Suppose the sites satisfy $|a_i - a_j| = r$ for all $i \neq j$, as for the vertices of a regular simplex. Then $N \leq d+1$. Put*

$$w = c_{d,s} m_1^2 r^{-\kappa}.$$

Then $\theta_1 = -(N-1)w$ and $\theta_2 = \cdots = \theta_N = w$. Thus one collective level is lowered by $(N-1)wL^{-\kappa}$, while the remaining $N-1$ levels are raised by $wL^{-\kappa}$, up to the remainder in (4.13).

Proof. After translating a_N to the origin, the $N-1$ vectors $a_i - a_N$ have Gram matrix with diagonal entries r^2 and off-diagonal entries $r^2/2$. This matrix is positive definite, so $N-1 \leq d$. Here $M_{\mathbf{a}} = -w(J-I)$, where J is the all-ones matrix. The vector $(1,\dots,1)$ has eigenvalue $-(N-1)w$, and every vector in its orthogonal complement is an eigenvector with eigenvalue w. ■

5. Numerical illustrations on intervals

We illustrate the fixed-mesh cluster asymptotic of Corollary 5.2 using conforming Galerkin matrices assembled from exact cell integrals. The mesh is held fixed as L increases; no discretization-error estimate as $h \to 0$ is asserted. Take $d = 1$, $0 < s < 1/2$, and consider finite unions of unit intervals. For high-index eigenvalue asymptotics and eigenfunction estimates on a single interval, see [Kwa12, Theorem 1 and Propositions 1–2]; sharper eigenvalue asymptotics and an eigenfunction bound uniform in both the index and the fractional order are proved in [Zha26, Theorems 1 and 2]. Both sources parameterize the operator order as $\alpha = 2s$.

Partition each component into cells of length h. The cell indicators have finite form energy by Proposition 5.1, so they span a conforming finite-dimensional subspace of the form domain.

Proposition 5.1 *(Exact cell-integral matrix)*. *Assume $0 < s < 1/2$. Let $I_1, \dots, I_M$ be pairwise disjoint cells of common length h, with centers x_α, and set $G = \bigcup_{\alpha=1}^M I_\alpha$, $r_{\alpha\beta} = |x_\alpha - x_\beta|$, and $q = 1-2s$. In the $L^2(G)$-orthonormal basis $e_\alpha = h^{-1/2}\mathbf{1}_{I_\alpha}$, the Galerkin matrix H_h of the zero-exterior form $\mathcal{Q}_G$ has entries*

$$(H_h)_{\alpha\alpha} = \frac{c_{1,s}}{h}\frac{2h^q}{2sq}, \tag{5.1}$$

$$(H_h)_{\alpha\beta} = -\frac{c_{1,s}}{h}\frac{2r_{\alpha\beta}^q - (r_{\alpha\beta}+h)^q - (r_{\alpha\beta}-h)^q}{2sq}, \qquad \alpha \neq \beta. \tag{5.2}$$

Proof. For two disjoint cells with center distance $r \geq h$, two direct integrations give

$$\iint_{I_\alpha \times I_\beta} |x-y|^{-1-2s}\,dx\,dy = \frac{2r^q - (r+h)^q - (r-h)^q}{2sq}.$$

Multiplication by $-c_{1,s}/h$ gives (5.2). Similarly, the diagonal entry in (5.1) is $c_{1,s}/h$ times

$$\int_{I_\alpha}\int_{\mathbb{R}\setminus I_\alpha} |x-y|^{-1-2s}\,dy\,dx = \frac{2h^q}{2sq}.$$ ■

Corollary 5.2 *(Fixed-mesh cluster asymptotic)*. *Let $D = (-1/2, 1/2)$, and fix a uniform partition of D. Let $\mu_{1,h}$ and $\phi_{1,h}$ be the lowest eigenvalue and a positive, L^2-normalized piecewise-constant eigenfunction of the one-well Galerkin matrix. Put $m_{1,h} = \int_D \phi_{1,h}$. Translate this same partition to each component $D + La_j$, and write the ordered eigenvalues of the resulting Galerkin matrix on $\bigcup_{j=1}^N (D + La_j)$ as $\lambda_{k,h}(L)$. Then, as $L \to \infty$, for fixed distinct $a_1, \dots, a_N$,*

$$\lambda_{k,h}(L) = \mu_{1,h} + L^{-1-2s}\theta_{k,h} + O_{h,\mathbf{a},s}\big(L^{-3-2s} + L^{-2-4s}\big), \qquad 1 \le k \le N, \tag{5.3}$$

where $\theta_{1,h} \le \cdots \le \theta_{N,h}$ are the eigenvalues of the matrix with zero diagonal and off-diagonal entries $-c_{1,s} m_{1,h}^2 |a_i - a_j|^{-1-2s}$.

Proof. All off-diagonal entries of the one-well matrix in (5.2) are negative. The Rayleigh-quotient argument in the proof of Corollary 4.9 therefore gives a simple lowest eigenvalue and a strictly positive eigenvector. On the direct sum of N fixed Galerkin spaces, the block expansion used in Lemma 4.2 is exact, its off-diagonal norm is $O_{h,\mathbf{a},s}(L^{-1-2s})$, and the complement of the N copied ground states is separated by the positive discrete one-well gap. The Taylor argument in Lemma 4.5, now in Euclidean operator norm, gives an $O_{h,\mathbf{a},s}(L^{-3-2s})$ compression error. Applying Lemma 4.3 to this finite matrix gives the additional $O_{h,\mathbf{a},s}(L^{-2-4s})$ term and proves (5.3). ∎

For the computation we take $s = 1/4$, $D = (-1/2, 1/2)$, and the standard normalization

$$c_{1,s} = \frac{4^s s\,\Gamma(1/2 + s)}{\sqrt{\pi}\,\Gamma(1 - s)}.$$

Table 1 records the one-well quantities, including the dependence on mesh size of $m_{1,h}$, which enters the interaction coefficient.

cells	$\mu_{1,h}$	$\mu_{2,h} - \mu_{1,h}$	$m_{1,h}$
60	1.379455	0.910554	0.962839
120	1.375369	0.899888	0.963351
180	1.374129	0.896965	0.963602

Table 1. One-well Galerkin data for $D = (-1/2, 1/2)$, $s = 1/4$, and the indicated uniform meshes.

On the 180-cell-per-component mesh, the two-well component centers are $(-L/2, L/2)$, while the three-well centers are $(-L, 0, L)$. For the corresponding fixed sites, the effective matrices on this mesh have eigenvalues

$$\begin{aligned} \mathbf{a} &= (-1/2, 1/2) : \quad (-0.18521477, 0.18521477), \\ \mathbf{a} &= (-1, 0, 1) : \quad (-0.29671332, 0.06548331, 0.23123001). \end{aligned}$$

Table 2 lists the scaled shifts computed from the Galerkin eigenvalues. For each configuration, the last column is

$$E_h(L) = \max_{1 \le k \le N} \big| L^{3/2}\big(\lambda_{k,h}(L) - \mu_{1,h}\big) - \theta_{k,h} \big|. \tag{5.4}$$

These errors are computed from the full-precision values before the shifts and effective spectra are rounded for display.

At the selected separations, the scaled errors decrease monotonically; Corollary 5.2 proves that they tend to zero at fixed mesh. Here $g_h = \mu_{2,h} - \mu_{1,h}$ and $R = 1/2$. At $L = 3$, the bound γ_L in (4.4) is 0.10858 for two wells and 0.21716 for three wells, while $g_h/4 = 0.22424$. Since γ_L decreases with L and $L\delta_{\mathbf{a}} \geq 4R$ already at $L = 3$, the finite-dimensional cluster estimate used in the proof applies at every computed separation. The calculation uses the exact entries (5.1)–(5.2); no singular quadrature or fitted coefficient enters the tables.

Two wells

L	$L^{3/2}(\lambda_{1,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{2,h} - \mu_{1,h})$	$E_h(L)$
3	−0.190740	0.189832	5.52×10^{-3}
6	−0.186544	0.186335	1.33×10^{-3}
12	−0.185551	0.185486	3.36×10^{-4}
24	−0.185302	0.185279	8.68×10^{-5}

Three wells

L	$L^{3/2}(\lambda_{1,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{2,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{3,h} - \mu_{1,h})$	$E_h(L)$
3	−0.304984	0.065562	0.237544	8.27×10^{-3}
6	−0.298738	0.065554	0.232747	2.03×10^{-3}
12	−0.297236	0.065504	0.231595	5.22×10^{-4}
24	−0.296852	0.065488	0.231316	1.38×10^{-4}

Table 2. Scaled cluster shifts for $s = 1/4$ and $D = (-1/2, 1/2)$ on the 180-cell-per-component mesh, with two wells centered at $(-L/2, L/2)$ or three wells centered at $(-L, 0, L)$. The maximum scaled-shift error $E_h(L)$ is defined in (5.4).

Use of artificial intelligence

The author used OpenAI's GPT-5.6 Sol model (`gpt-5.6-sol`) to explore and test arguments, audit citations and calculations, prepare reproducibility and release-checking code, and revise the manuscript. The author checked the final arguments and takes responsibility for all mathematical claims, citations, computations, and wording.

Code availability

The script `check_asymptotics.py` reproduces Tables 1 and 2. The ancillary files also include the pinned dependencies in `requirements-numerics.txt` and the reference output in `asymptotics_180.txt`. The calculation was reproduced using Python 3.12.3, NumPy 1.26.4, and SciPy 1.11.4. These files are archived with the manuscript and are also available, with the manuscript source, in skypher/laplace-tunneling, tag paper-2026-09-05-r2.

References

[AFN20] Laura Abatangelo, Veronica Felli, and Benedetta Noris. On simple eigenvalues of the fractional Laplacian under removal of small fractional capacity sets. *Communications in Contemporary Mathematics*, 22(8):1950071, 2020. doi:10.1142/S0219199719500718; arXiv:1902.03550v2.

[BDVV23] Stefano Biagi, Serena Dipierro, Enrico Valdinoci, and Eugenio Vecchi. A Hong–Krahn–Szegö inequality for mixed local and nonlocal operators. *Mathematics in Engineering*, 5(1):1–25, 2023. doi:10.3934/mine.2023014; arXiv:2110.07129.

[BP16] Lorenzo Brasco and Enea Parini. The second eigenvalue of the fractional p-Laplacian. *Advances in Calculus of Variations*, 9(4):323–355, 2016. doi:10.1515/acv-2015-0007; arXiv:1409.6284v2.

[DFW21] Sidy Moctar Djitte, Mouhamed Moustapha Fall, and Tobias Weth. A fractional Hadamard formula and applications. *Calculus of Variations and Partial Differential Equations*, 60(6):231, 2021. doi:10.1007/s00526-021-02094-3; arXiv:2002.07719.

[DNPV12] Eleonora Di Nezza, Giampiero Palatucci, and Enrico Valdinoci. Hitchhiker's guide to the fractional Sobolev spaces. *Bulletin des Sciences Mathématiques*, 136(5):521–573, 2012. doi:10.1016/j.bulsci.2011.12.004; arXiv:1104.4345.

[DPLSV26] Serena Dipierro, Edoardo Proietti Lippi, Caterina Sportelli, and Enrico Valdinoci. Nonlocal eigenvalue problems and superposition operators, 2026. arXiv:2602.18035v1 [math.AP], 20 February 2026.

[GS19] Divya Goel and K. Sreenadh. On the second eigenvalue of combination between local and nonlocal p-Laplacian. *Proceedings of the American Mathematical Society*, 147(10):4315–4327, 2019. doi:10.1090/proc/14542; arXiv:1901.03444.

[Kwa12] Mateusz Kwaśnicki. Eigenvalues of the fractional Laplace operator in the interval. *Journal of Functional Analysis*, 262(5):2379–2402, 2012. doi:10.1016/j.jfa.2011.12.004; arXiv:1012.1133.

[LR23] Alexis Léculier and Jean-Michel Roquejoffre. Some effects of nonlocal diffusion on the solutions of Fisher–KPP equations in disconnected domains. *Calculus of Variations and Partial Differential Equations*, 62(1):30, 2023. doi:10.1007/s00526-022-02374-6; arXiv:2004.14771.

[PS20] Enea Parini and Ariel Salort. Compactness and dichotomy in nonlocal shape optimization. *Mathematische Nachrichten*, 293(11):2208–2232, 2020. doi:10.1002/mana.201800234; arXiv:1806.01165.

[Wu26] Zijian Wu. Equilibrium configurations and stability analysis of a discrete toy model arising from the minimization of eigenvalues of the fractional Laplacian. *Applied Mathematics and Computation*, 531:130195, 2026. doi:10.1016/j.amc.2026.130195.

[Zah26] Alvis Zahl. Minimizing eigenvalues of the fractional Laplacian. *Calculus of Variations and Partial Differential Equations*, 65(3):98, 2026. doi:10.1007/s00526-026-03270-z; arXiv:2504.09840.

[Zha26] Cheng Zhang. Eigenvalue asymptotics and uniform eigenfunction bounds for the fractional Laplacian in the interval, 2026. arXiv:2608.23457v1 [math.CA], 24 August 2026.